
UNIT 2 THE KEYNESIAN MODEL *

Structure

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20.0 OBJECTIVES

After going through this Unit you should be in a position to

- appreciate the assumptions underlying the Keynesian model;
- explain the concepts of consumption function and saving function;
- identify the factors affecting investment demand;
- show that equilibrium output in the Keynesian model is determined by the level of aggregate demand;
- explain the various types multipliers in the Keynesian system;
- explain the paradox of thrift;
- explain the concepts of liquidity preference and liquidity trap; and
- discuss the Keynesian view on the role of the government in the economy .

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2.1 INTRODUCTION

In this Unit we will learn about the Keynesian model that was first developed by the British economist John Maynard Keynes in his famous book ‘The General Theory of Employment, Interest and Money’, published in 1936. This model differs in several respects from the classical approach to macroeconomics that you studied in the previous Unit. The Keynesian model allows for the existence of unemployment as well as rigidity in wages and prices. It recognizes that lack of effective demand may prevent the attainment of full employment of resources. Also, it emphasizes the need for government intervention – as the market mechanism would not be able to restore full employment equilibrium due to rigidities in wage rate and prices.

2.2 COMPONENTS OF AGGREGATE DEMAND

We observe that Keynesian economics gained importance against the backdrop of the Great Depression of the 1930s, when developed economies like the USA and Great Britain faced a deep and prolonged recession, with rising unemployment, falling revenues and profits of firms and rise in unutilized capacity. In this situation the market mechanism failed to restore equilibrium and the Keynesian policy of raising aggregate demand via increase in public expenditure played a crucial role in addressing the problem.

Aggregate demand is the planned expenditure on goods and services by all the economic agents, such as households, firms and government. There are three main components of aggregate demand in a closed economy: (i) *planned* consumption expenditure (C) of households, (ii) *planned* investment expenditure (I) of firms, and (iii) expenditure incurred by the government on goods and services (G). Throughout our discussion in this Unit, each of C, I and G are assumed to be in real terms (i.e., no change in prices).

2.2.1 Consumption Function

Keynes proposed the idea of a Consumption function which gives the relation between *planned* consumption expenditure of households (C) and aggregate income (Y). It can be represented by the following linear equation:

$$C = a + bY \quad \dots (2.1a)$$

In the presence of the government, a part of income would be paid in taxes (T). In that case the consumption function would be represented as a function of disposable incomes ($Y_d = Y - T$), that is,

$$C = a + b(Y - T) = a + bY_d \quad \dots (2.1b)$$

For the discussion below, for simplicity, we assume that there is no government, so that $T = 0$ and $Y = Y_d$.

The slope of the consumption function ‘ b ’ is assumed to be a positive fraction (i.e., $0 < b < 1$). It means that the consumption function can be represented as an

upward sloping straight line as in Fig. 2.1. It indicates that planned consumption expenditure increases with an increase in the level of aggregate income.

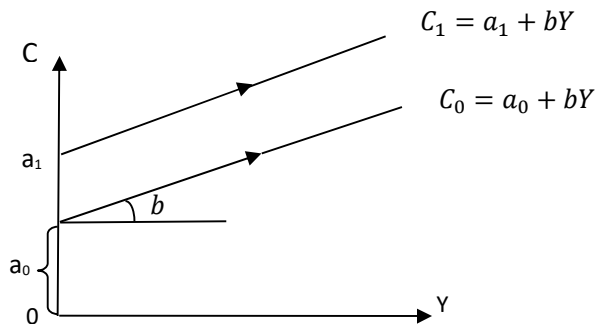


Fig. 2.1: Consumption Function

Note that ‘ b ’ is defined as the *marginal propensity to consume* (MPC) and it represents the change in consumption induced by a unit change in income. The MPC is assumed to be less than one. If there is an increment in income by ΔY , the induced increase in consumption ΔC is a fraction of ΔY (i.e., $\Delta C = b\Delta Y$ and $\Delta C/\Delta Y = b < 1$). It means that the entire increment in income is **not** spent on consumption, a fraction $(1 - b)$ is saved. It means total saving out of the incremental income is $(1 - b)\Delta Y$.

Let us consider the case where the consumption function is given by $C = 20 + 0.25Y$. If aggregate income increases by Rs.100 (i.e., $\Delta Y = 100$), then Rs. 25 ($\Delta C = 0.25 \times 100$) is the additional consumption and Rs.75 $[(1 - 0.25) \times 100]$ is additional saving of households out of the incremental income. Similarly, if income declines by Rs.100, then consumption would decline by Rs. 25 only, while saving would decline by Rs. 75.

The intercept of the consumption function ‘ a ’ represents autonomous consumption, with $a > 0$. From equation (2.1a), you can see that when $Y = 0$, we have $C = a$. It implies that consumption occurs even when income is zero. This consumption is financed by drawing down past saving or by ‘dis-saving’. An increase in ‘ a ’ represents increase in planned consumption of households at each level of income; this can happen, for instance, when consumer confidence increases (i.e., households’ expectations about the future turn buoyant). You can see in Fig. 2.1 that an increase in the intercept (from a_0 to a_1) results in a parallel, upward shift of the consumption function, from C_0 to C_1 .

Aggregate consumption has two components: (i) an autonomous component ‘ a ’ that does not depend on income, and (ii) an ‘induced’ component ‘ bY ’ that depends on aggregate income levels. An increase (decrease) in Y induces an increase (decrease) in planned consumption expenditure by households.

The ratio of aggregate consumption expenditure to aggregate income $\left(\frac{C}{Y}\right)$ is defined as the *average propensity to consume* (APC). From equation (2.1a) you

can see that $APC = C/Y = a/Y + b$. Note that APC will decline as Y increases, while MPC remains constant.

2.2.2 Investment Function

The investment function (I) represents *planned* investment expenditure undertaken by firms on machinery, factories and other such items that add to productive capacity of the economy. In the simplest version of the Keynesian model investment is assumed to be ‘autonomous’, i.e., independent of the level of aggregate income (Y). In this case the investment function is represented by the equation as below.

$$I = \bar{I} \quad \dots (2.2a)$$

In equation (2.2a) a ‘bar’ over the variable investment ($\bar{I}$) variable indicates a fixed amount. However, we can easily relax this assumption and allow investment to be a function of aggregate output. In this case the investment function is:

$$I = c + dY \quad \dots (2.2b)$$

where $c > 0$ and $d > 0$; ‘ c ’ is the autonomous part of investment, d is the *marginal propensity to invest* (MPI) and ‘ dY ’ represents investment that is induced by aggregate income.

Fig. 2a below depicts an autonomous investment function, while Fig. 2.2b depicts an induced investment function which has a component that is induced by income.



Fig. 2.2a: Autonomous Investment

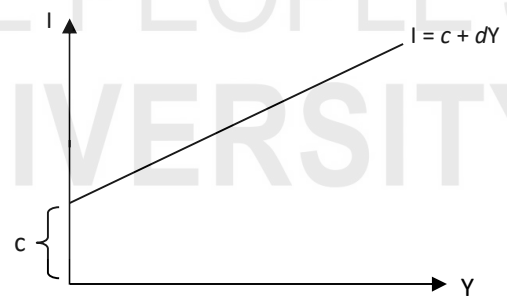


Fig. 2.2b: Induced Investment

Note that the term investment refers to planned expenditure by firms on new production equipment (e.g., machinery, building, etc.) that add to productive capacity. It does *not* refer to buying of corporate bonds or shares in a company or opening a fixed deposit account in a bank.

An important variable that affects firms’ decision to invest is the rate of interest. As the rate of interest increases, the cost of borrowing goes up, resulting in a reduction in the firms’ planned investment expenditure. The inverse relation between investment and interest rate can be further explained as follows:

Say, the current interest rate is i and an investment project that costs Rs. A , is expected to last for T periods. In each period it yields a return of R_t , where $t = 1, 2, \dots, T$. In this case, the present value of the stream of returns from the project P is calculated as:

$$P = \frac{R_1}{(1+i)} + \frac{R_2}{(1+i)^2} + \frac{R_3}{(1+i)^3} + \dots + \frac{R_T}{(1+i)^T} \quad \dots (2.2c)$$

If profit is the only motive of investment, this project will be taken up only if $(P - A) \geq 0$.

For example, if you are willing to lend Rs.100 at 10 per cent rate of interest for a period of 1 year, then at the end of the year you will receive Rs.110. So, at the current interest rate of 10 per cent, the *present discounted value* of Rs.110 is Rs. 100 [where, $100 = 110 / (1+0.1)$].

Now, for the economy as a whole, for any given rate of interest, i_0 , the corresponding level of investment I_0 by firms would consist of all the projects for which $P \geq A$. Now, if the interest rate *increases* to i_1 (R_t and A remaining unchanged), P would *decrease*. Consequently, some projects for which $P \geq A$, would now be unprofitable. At a higher interest rate, therefore, all the projects for which $P < A$ would not be taken up, so there would be a fall in the associated level of investment I_1 . Thus, when $i_0 > i_1$ we have $I_0 < I_1$.

If we include interest rate as an explanatory variable in the investment function, we have

$$I = c + dY - e i \quad \dots (2.2d)$$

where $c > 0$, $d > 0$ and $e > 0$. Here, e captures the interest responsiveness of planned investment expenditure.

Check Your Progress 1

- 1) Discuss the main features of the Keynesian consumption function.

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- 2) Through a diagram, explain how (i) a decrease in autonomous consumption, and (ii) an increase in the MPC, would affect the consumption function.

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- 3) Distinguish between autonomous investment and induced investment.

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- 4) State the reasons for the inverse relationship between interest rate and investment level.

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2.3 DETERMINATION OF OUTPUT IN THE KEYNESIAN MODEL

The Keynesian model is based on certain assumptions. First, there are unemployed resources in the economy. In other words, there is underutilized productive capacity (e.g., factories operating below capacity, unused machines, labour without employment, etc.). It follows from this that output in such an economy is demand determined, i.e., existing production capacity can be used to step up production in case there is an increase in aggregate demand for output. Second, money wages and prices are not flexible (i.e., there are price rigidity and wage rigidity). Since there is unemployment and workers are willing to work at the going wage rate, an increase in the supply of output is possible at the given price level. Third, output is assumed to be a homogenous commodity used both for consumption and investment.

We will present the simplest Keynesian model applicable to a three-sector economy consisting of households, firms and the government. The economy is closed (i.e., no external trade). Later on we relax this assumption and allow for external trade.

In the three-sector economy, aggregate demand comprises (i) planned consumption by households (C), (ii) planned investment by firms (I), and (iii) expenditure on goods and services by the government (G). Initially we assume that investment is fully autonomous, an assumption that is relaxed later. The government is assumed to raise taxes and here we consider the cases of both lump sum taxes as well as proportional taxes. Later we will also consider the implications that arise when the government makes transfer payments to households (e.g., pensions, unemployment dole, etc.).

Therefore, in our simple Keynesian model

$$\text{Aggregate Demand (AD)} = C + I + G$$

By substituting the values of C, I and G, we obtain

$$AD = a + b(Y - T) + \bar{I} + G \quad \dots (2.3)$$

Here T is a lump sum tax levied by the government; the investment function is given by (2.2a).

2.3.1 Determination of Equilibrium Output

The equilibrium condition in the Keynesian model is that aggregate demand for output is equal to aggregate supply (Y) or, the total output produced in the economy. Note that at equilibrium, the plans of households and firms (regarding expenditure on consumption and investment) are also fulfilled. In case plans are *not* fulfilled there would be changes in aggregate demand (as households and firms would adjust their plans) and hence output produced would change as well, so that the situation would *not* be one of equilibrium.

The equilibrium condition in the economy is achieved when

$$\text{Aggregate Supply} = \text{Aggregate Demand}$$

From equation (2.3) we know that

$$Y = a + b(Y - T) + \bar{I} + G$$

Solving this for Y, equilibrium output (Y_e) can be obtained as:

$$Y = a + bY - bT + \bar{I} + G$$

Or,

$$Y - bY = a - bT + \bar{I} + G$$

Thus,

$$Y_e = (a - bT + \bar{I} + G)/(1 - b) \quad \dots (2.4)$$

From (2.4) it should be clear to you that equilibrium output would be higher at higher levels of (i) autonomous investment, (ii) government expenditure, and (iii) the MPC. Recall that $0 < \text{MPC} < 1$, so that the term $(1 - b)$ in the denominator is always positive.

We will now carry out graphical analysis to further understand the process of attainment of equilibrium in the simple Keynesian model. First we will consider the case of a two-sector economy with only households and firms and no government. Later we will consider a three-sector economy which has the government sector as well.

Two-Sector Economy without Government

In a two-sector economy, there are only households and firms. Therefore, aggregate demand has only two components, C and I. Thus, $AD = C + \bar{I}$. Setting $G = T = 0$ in equation (2.4) above, we see that equilibrium output would be

$$Y_e = (a + \bar{I}) / (1 - b) \quad \dots (2.5)$$

We depict equilibrium level of output in Fig. 2.3. Along the horizontal axis we measure output or income (recall that income is generated in the process of production of output, so we use the terms income and output interchangeably). Along the vertical axis we measure aggregate demand and its components (C and I).

Note that OL is a line which makes an angle of 45° with the horizontal axis, so that a perpendicular from this line such as AY_1 , forms an isosceles triangle AOY_1 , with $\angle OAY_1 = \angle AOY_1 = 45^\circ$, and as per the property of isosceles triangles, the two sides OY_1 and AY_1 are equal. Using this construct we are able to compare vertical distances (demand) with the horizontal distances (output or income). Note that at A , where the consumption function cuts OL , consumption expenditure AY_1 is exactly equal to income OY_1 , so that saving is zero.

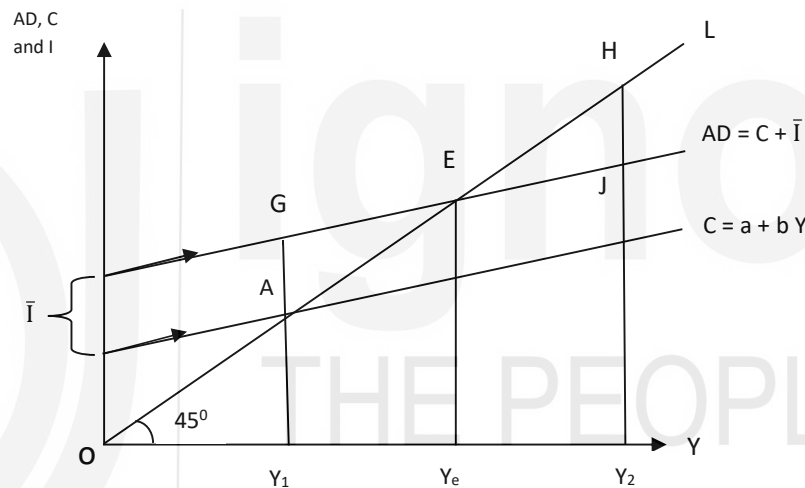


Fig. 2.3: Equilibrium Output in a Two-Sector Economy

In Fig. 2.3 we derive the aggregate demand curve (AD) by adding the investment ($\bar{I}$) and consumption (C) functions. Since investment is autonomous the line AD is parallel to the consumption function, i.e., slope of AD is the same as the slope of the consumption function. Equilibrium output is attained at E , at the intersection of AD with OL where aggregate demand EY_e is equal to aggregate output OY_e . Note that the equilibrium is unique, as there is only one level of income OY_e at which aggregate demand is equal to aggregate supply. In this case firms will be able to sell exactly as much as they had planned and hence there will be no unplanned changes in their stock of inventories.

For all income levels less than OY_e , i.e., to the left of E , there is excess demand for output. For example, when aggregate income is $OY_1 (=AY_1)$, the corresponding level of aggregate demand GY_1 is greater than the supply of output $AY_1 (=OY_1)$. In this case since demand is more than the output produced by firms, they will have to run down their stock of inventories. So in the next

period, firms will increase production and output would move toward OY_e . This process of increase in output (and hence income) will continue till OY_e is reached, where the amount produced (OY_e) being exactly equal to the amount demanded (EY_e), there is no unplanned changes in the stock of inventories and hence no further change in production levels.

Similarly, to the right of E (i.e., at all income levels greater than OY_e), there will be excess supply of output. When income is OY_2 , the corresponding level of aggregate demand JY_2 is less than the supply $HY_2 (= OY_2)$. In this case, firms will be unable to sell the entire amount produced ($HY_2 > JY_2$) and hence their stock of inventories will increase (i.e., there will be unplanned inventory accumulation by firms). This would induce firms to reduce production in the next period. This process will continue till the equilibrium output $EY_e (= OY_e)$ is reached, where demand and supply are equal and there are no more unanticipated changes in firms' stock of inventories.

From this discussion we see that output market equilibrium in the Keynesian model is stable. Any situation of disequilibrium tends to be self-correcting, inducing a movement back towards the equilibrium level of output.

Three-Sector Economy with Government

In equation (2.4) we derived equilibrium output in a three sector model, with households, firms and a government sector. Diagrammatically this can be shown in Fig. 2.4. As earlier, the OL line makes an angle of 45° with the horizontal axis along which we measure income (or output). Aggregate demand is measured along the vertical axis. Now, this has three components, C, I and G, where investment and government expenditure are autonomous.

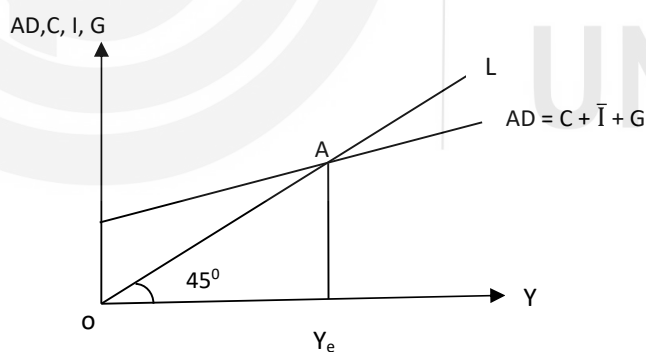


Fig. 2.4: Equilibrium Output in a Three-sector Economy

Equilibrium in the three-sector model occurs at the intersection of the AD curve and the OL line, where demand AY_e is equal to aggregate supply OY_e . You should see that equilibrium income level OY_e is unique and that this equilibrium is stable. At all income levels higher than OY_e there is excess supply, so that firms decrease production, inducing a change in output towards OY_e . On the other hand, at all income levels lower than OY_e , there is excess demand, inducing firms to increase production.

Note that the position of the AD curve is determined by the size of the autonomous components of aggregate demand (viz., I , G and autonomous consumption a). The higher the levels of these components, the higher would be the AD line and the higher would be the equilibrium output level (Y_e), implying that output is demand-determined in the Keynesian model.

2.3.2 Investment Multiplier

We will now examine the comparative static properties of the Keynesian model and explain the concept of the multiplier.

Autonomous Investment

Consider once again the two-sector model with only firms and households, such that equilibrium income is $Y_e = (a + \bar{I})/(1 - b)$. If, *ceteris paribus*, there is a change in the level of autonomous investment (ΔI), the associated change in income would be given by $\Delta Y = (1/(1 - b) \times \Delta I$. Thus, $\frac{\Delta Y}{\Delta I} = (1/(1 - b))$. Here, we consider change in one component of autonomous demand at a time. This means when $\Delta I > 0$, we shall assume that $\Delta a = 0$ and so on. The term $(\frac{1}{1-b})$ is greater than one, since our assumption is that $(0 < b < 1)$. An implication of the above is that when investment increases by ΔI , the associated increase in income is greater than ΔI , as it is a multiple of ΔI . How is this possible? To see this, consider the several rounds of increments in expenditure that are triggered by a onetime increase in autonomous expenditure.

If we increase I by ΔI , income will increase by ΔI in the first round, which will induce a second round increase in consumption by $b\Delta I$ (since consumption is a function of income). Since output is demand-determined, this increase in consumption demand raises output and hence income increases further by $b\Delta I$. This induces a third round increase in consumption by $[b(b\Delta I) = b^2\Delta I]$ which triggers further increases in income ($b^2\Delta I$). In the fourth round, consumption demand increases by $[b(b^2\Delta I) = b^3\Delta I]$. The process continues with increase in consumption raising output, income and yet another round of consumption. Note that since $MPC < 1$ (i.e., $b < 1$), each subsequent round of increase in consumption is weaker than the previous one, so that the entire process dies down gradually.

Therefore, the total increase in income due to a onetime increase in autonomous investment is:

$$\Delta Y = (\Delta I + b\Delta I + b^2\Delta I + b^3\Delta I + \dots) = \Delta I (1 + b + b^2 + b^3 + \dots)$$

...(2.6) Since $0 < b < 1$, the expression in parentheses on the right hand side of (2.6) can be expressed as the sum of a convergent infinite geometric progression series, i.e., $1 + b + b^2 + b^3 + \dots = 1/(1 - b)$. Therefore, $\Delta Y = [1 / (1 - b)] \Delta I$, which shows that income increases by a multiple of the initial increase in autonomous investment and the multiplier is $(\frac{1}{1-b})$. This is shown diagrammatically in Fig. 2.5.